

PREPARED FOR: TREVOR KNIGHT, GMD · WILMINGTON CAMPUS STRATEGY

Merge4Ward + M4CleanTech

Wilmington Clean Industrial Campus

A co-located manufacturing and clean-tech strategy for the Hay Road Peninsula, Wilmington DE — using a single City ground lease to simultaneously establish the Forward Building System's Mid-Atlantic manufacturing node, a M4CleanTech DE pyrolysis facility, and a procurement leverage platform that reduces Merge4Ward's per-panel material cost by an estimated 12–22% vs. a standard purchase model.

Merge 4 Design LLC · M4Di™ · Merge4Ward LLC · M4CleanTech LLC · Aubyn Architecture LLC

12+ ac

Available peninsula parcels
— Hay Road / Christina
River, Wilmington DE

\$0

Acquisition capital required
— City ground lease
eliminates purchase
requirement

12–22%

Estimated ROM reduction in
per-panel material cost
unlocked by confirmed site
control

2 nodes

Simultaneous DE + PA
deployment: Wilmington
(Merge4Ward +
M4CleanTech) + Chester
(M4CleanTech BOO)

SECTION 1 — THE SITE: HAY ROAD PENINSULA, WILMINGTON DE

Why This Peninsula. Why Now.

The Hay Road Peninsula is a horseshoe-shaped industrial promontory where the Christina River wraps around three sides before meeting the Delaware River. It sits within the City of Wilmington's jurisdiction, inside the Port of Wilmington logistics corridor, and is currently underutilized — a mix of brownfield land, light industrial operations, and vacant parcels carrying 26-series New Castle County parcel IDs. More than 12 acres of available or underproductive parcels are visible across the peninsula.

MERGE4WARD MANUFACTURING NODE	M4CLEANTECH DE FACILITY	SHARED INFRASTRUCTURE
Forward Building System — CLT/SIP Panel Production Western cluster parcels (26-044-00-014 / -018): 4–6 acres. Existing industrial infrastructure, heavy truck access, potential rail spur. 30,000–50,000 SF manufacturing building + panel staging laydown yard. Ships CLT/SIP panels to DE, MD, PA, and	Klean/KBR TyreNova BOO — Tire Pyrolysis Southern tip parcel (26-052-00-004): 3–5 acres. Three-sided river frontage enables barge offtake for pyrolytic oil, recovered carbon black, and steel — cutting product distribution cost vs. inland sites. Delaware node in the Mid-Atlantic	Rail Spur + Utilities + Laydown Central rail yard parcel (26-044-00-021): 2–3 acres. If active CSX/NS spur confirmed, inbound CLT lumber and outbound panel distribution by rail. Shared utility connections, security perimeter, and truck staging serve both Merge4Ward and M4CleanTech —

NJ projects at lower trucking cost than any competing inland site.

Target: 26-044 western cluster

M4CleanTech network, running parallel to Chester PA.

Target: 26-052 southern tip

splitting infrastructure cost between two tenants.

Target: 26-044-00-021 rail yard

WHY THE CITY OF WILMINGTON SAYS YES

The City gets: 40–80 Merge4Ward manufacturing jobs + 25–40 M4CleanTech operations jobs + brownfield remediation of a peninsula it currently cannot activate + Opportunity Zone investment trigger + new tax ratables. M4D's existing relationship with Sean Park (OED) means this is a deepening conversation, not a cold ask. One master development agreement covers the entire peninsula — the City partners on the whole vision or it doesn't.

SECTION 2 — THE PROCUREMENT LEVERAGE THESIS

Site Control Changes Every Supplier Conversation

This is the argument Trevor needs to understand before any supplier meeting. A signed long-term ground lease with the City of Wilmington is not just a real estate instrument — it is a negotiating credential that transforms Merge4Ward from a startup buyer into a committed, site-locked manufacturer with a defined production ramp. Every supplier of structural building components prices their product based on perceived buyer commitment. Confirmed site control signals permanence. Permanence unlocks pricing, terms, and exclusivity arrangements that are simply unavailable to a development-stage buyer without a facility.

The mechanism works on three levels:

- ▶ **Volume commitment credibility.** A manufacturer with a permitted, ground-leased facility can credibly commit to annual purchase volumes. A developer without site control cannot. Suppliers price uncertainty — when uncertainty disappears, margin compression follows.
- ▶ **Exclusivity leverage.** For suppliers seeking a Mid-Atlantic distribution point — TerraLam, Structurlam, AAC manufacturers, Thermo-Brace, Framecrete — a confirmed Wilmington facility is a potential regional anchor relationship, not a one-time order. Anchor relationships carry preferred pricing, supply priority, and co-marketing value that spot buyers never access.
- ▶ **Financing instrument effect.** A ground lease is a bankable instrument. Suppliers with vendor financing programs — and several CLT manufacturers offer this — will extend more favorable terms when the buyer has a real estate position that demonstrates solvency and long-term operating intent.

SUPPLIER / SYSTEM	WITHOUT GROUND LEASE	WITH CONFIRMED GROUND LEASE	ROM COST DELTA
CLT Panel Supply (TerraLam /	Spot buyer: list price \$180–\$240/m ³ . No volume discount. 12–16 week lead	Anchor manufacturer relationship: \$145–\$195/m ³ at contracted annual volume.	~\$35–\$55/m ³ savings (18–23% reduction)

SUPPLIER / SYSTEM	WITHOUT GROUND LEASE	WITH CONFIRMED GROUND LEASE	ROM COST DELTA
Structurlam / Mercer Mass Timber)	times. No supply priority. No vendor financing.	6–8 week lead times. Supply allocation priority. Potential vendor financing on first facility order.	
SIP Panel Supply (Acme Panel / Premier SIPs / Panelwrights)	Project-by-project procurement. Full list price. No standing order discount. Fabrication queue position uncertain.	Standing annual supply agreement: 8–15% volume discount on committed annual SF. Dedicated fabrication queue slot. Co-branding as Mid-Atlantic distributor.	8–15% reduction on SIP panel cost
AAC Block / Panel (Aercon / Xella / Brickform)	Standard distributor pricing. Freight from nearest stocking warehouse adds \$15–\$30/panel equivalent.	Direct manufacturer supply agreement at Wilmington node. Freight savings (DE river port) + 10–18% volume pricing.	10–18% product + freight savings combined
Thermo-Brace Structural Insulated Sheathing	Broker/distributor pricing. No direct manufacturer relationship. No preferred access to limited supply.	Direct OEM supply agreement as regional anchor. Priority allocation of limited production runs. 10–15% below distributor.	10–15% below distributor + supply security
Framecrete / ICF Systems (Nudura / Fox Blocks / Logix)	Contractor pricing tier. No volume rebate. No regional distribution rights.	Regional distributor or preferred installer relationship: 12–20% below contractor tier + territory exclusivity for DE/PA/NJ/MD market.	12–20% cost reduction + territorial moat
AGGREGATE IMPACT (All systems, Yr 2–3 steady state)	Standard market pricing across all systems. No leverage. No supply security. Higher per-panel material cost.	Blended material cost reduction across CLT/SIP/AAC/sheathing/ICF with committed site and annual volume.	ROM 12–22% blended material cost reduction

ROM cost delta estimates based on published manufacturer volume discount schedules, regional freight savings from waterfront DE location, and comparable preferred supplier agreements in the CLT/mass timber market. Actual savings depend on negotiated terms, committed annual volumes, and product mix. Merchant review required before committing to any supply agreement.

SECTION 3 — FINANCIAL PICTURE: THE CAMPUS ECONOMICS

What the Wilmington Campus Generates — ROM

The Wilmington Campus operates as three revenue streams sharing one ground lease cost. Merge4Ward manufacturing revenue, M4CleanTech ground lease spread, and procurement savings (which reduce project cost and improve margin on every Forward Building System delivery) all flow from the same foundational instrument: the City ground lease.

Stream 1 — Merge4Ward Manufacturing Revenue

REVENUE STREAM	DESCRIPTION	ROM AMOUNT	TIMING
Panel Production Revenue	CLT/SIP/AAC panels fabricated and sold to M4D project pipeline + third-party developers in DE/PA/NJ/MD market. Year 3 steady state at 40–60% facility utilization.	\$4.2M–\$7.8M gross/yr	Year 2–3 ramp
Material Cost at Market	Without ground lease / supplier agreements — standard procurement, no volume discount.	(\$2.8M–\$5.6M)/yr	Baseline
Material Cost with Agreements	12–22% blended reduction from confirmed ground lease procurement leverage + direct supplier agreements.	(\$2.2M–\$4.4M)/yr	Post-agreement
Gross Margin Improvement	Procurement savings flow directly to Merge4Ward gross margin — not passed to buyer unless competitively necessary.	+\$600K–\$1.2M/yr	Recurring annual
Operating Costs (ex-material)	Labor (40–80 FTE), facility, utilities, insurance, debt service on equipment.	(\$800K–\$1.6M)/yr	Steady state
NET OPERATING INCOME — Merge4Ward	Manufacturing NOI at Year 3 steady state.	\$1.2M–\$1.6M/yr	Year 3+

Stream 2 — M4CleanTech DE Revenue

REVENUE STREAM	DESCRIPTION	ROM AMOUNT	TIMING
Ground Lease Spread	City charges M4CleanTech nominal rent (\$1–\$5K/yr). Klean/KBR BOO sublessee pays M4CleanTech market industrial ground rent (\$50K–\$120K/yr). M4CleanTech pockets spread.	\$45K–\$119K/yr	From BOO execution
Developer Fee (one-time)	M4CleanTech bills Klean/KBR for DE site development services under PSA. Recoverable at BOO financial close.	\$360K–\$600K	At BOO close

REVENUE STREAM	DESCRIPTION	ROM AMOUNT	TIMING
Retained Equity Stake	5–15% equity participation in DE BOO facility negotiated at deal formation.	5–15% facility value	Exit Year 8–15
DE M4CleanTech NOI Total (ground lease only)	Cash flowing from Day 1 of BOO operations.	\$45K–\$119K/yr	Year 1 operations

Stream 3 — Procurement Savings as Margin Enhancement

This stream is often undervalued in project finance because it doesn't appear as revenue — it appears as cost reduction. But for Merge4Ward, a 12–22% blended reduction in material cost on a \$4.2M–\$7.8M revenue base is equivalent to adding \$600K–\$1.2M in annual gross margin without selling a single additional panel. That margin flows to either enhanced project competitiveness (lower bid prices vs. conventional construction) or retained Merge4Ward profit — or both.

WITHOUT GROUND LEASE / SUPPLIER AGREEMENTS	WITH CONFIRMED GROUND LEASE + AGREEMENTS
Material cost: \$2.8M–\$5.6M/yr No volume discount — spot buyer pricing across all systems No supply security — queue position uncertain for every order No territorial advantage — any competitor bids against same pricing	Material cost: \$2.2M–\$4.4M/yr 12–22% blended cost reduction via committed volume + direct OEM agreements Dedicated supply allocation — fabrication queue priority for all systems Regional exclusivity potential for DE/PA/NJ/MD territory with key suppliers
Gross Margin at Market: 25–28%	Gross Margin with Agreements: 35–42%

SECTION 4 — CAMPUS AGGREGATE: WHAT THE GROUND LEASE UNLOCKS TOTAL

One Instrument. Three Revenue Streams. One City Conversation.

The Wilmington ground lease is a single negotiation with Sean Park / City of Wilmington OED that simultaneously activates all three revenue streams. The City's cost: nominal ground rent received plus the administrative overhead of one master development agreement. The City's benefit: 65–120 permanent jobs, brownfield activation, Opportunity Zone investment, and tax ratables on a peninsula that currently generates almost nothing.

COMPONENT	DESCRIPTION	ROM AMOUNT	TIMING
Merge4Ward Manufacturing NOI	Panel fabrication revenue minus material cost (with supply agreements) minus operating costs. Year 3 steady state.	\$1.2M–\$1.6M/yr	Year 2–3
M4CleanTech DE Ground Lease Spread	BOO sublease spread after paying nominal City rent. Cash flows from Day 1 of Klean/KBR operations.	\$45K–\$119K/yr	Year 1 ops
Procurement Margin Enhancement	Blended material cost savings flowing to Merge4Ward gross margin. Not shown as separate P&L line — embedded in manufacturing NOI above.	\$600K–\$1.2M/yr	Recurring
M4CleanTech Developer Fee (one-time)	PSA fee from Klean/KBR for DE site development services, recoverable at BOO financial close.	\$360K–\$600K	One-time
CAMPUS TOTAL — Steady State	Annual recurring revenue streams from all three campus components. Developer fee excluded (one-time).	\$1.25M–\$1.72M/yr	Year 2–3+
City of Wilmington OZ + Brownfields Capital	Opportunity Zone equity deferral + EPA Brownfields Assessment/Cleanup grants + DEDO manufacturing incentives. Non-dilutive capital reducing facility build cost.	\$1.5M–\$4.5M	One-time grants
Permanent Jobs Created — Campus Total	Merge4Ward: 40–80 manufacturing FTE. M4CleanTech DE: 25–40 operations FTE.	65–120 jobs	Year 2–4

ROM figures ±30–40% confidence. Manufacturing NOI includes procurement savings already embedded in the margin calculation. Procurement savings ROM based on published CLT/SIP/AAC manufacturer volume discount schedules and regional freight benchmarks. Actual results depend on negotiated supplier agreement terms and annual committed volumes.

SECTION 5 — FINANCE STACK: HOW THE CAMPUS GETS BUILT

Grant-Heavy, Low-Equity, Ground-Lease-Anchored

The Wilmington Campus financing strategy mirrors the Chester structure: outside capital does the heavy lifting, the ground lease eliminates acquisition cost, and M4CleanTech's BOO structure means Klean/KBR funds the clean tech facility entirely. Merge4Ward's manufacturing facility is the primary capital requirement — and that facility is the most grant-eligible manufacturing project in Delaware given its EJ location, job creation profile, and advanced manufacturing classification.

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
EPA Brownfields Assessment + Cleanup	Peninsula has documented brownfield history. EPA grants fund Phase I/II ESA + remediation. EJ community overlay strengthens application.	\$500K–\$1.5M	0% grant	Non-dilutive, non-recourse
DEDO — Delaware Economic Development	DEDO Strategic Fund + Brownfield Assistance Program. Manufacturing job creation in Wilmington is core DEDO mandate. Sean Park / OED are the introduction pathway.	\$500K–\$2M	0% grant	Non-dilutive
EDA Economic Adjustment Assistance	Wilmington qualifies as economically distressed community. Manufacturing job creation + site remediation are EAA-eligible. Chester PA application establishes track record.	\$1M–\$3M	0% grant	Non-dilutive
Opportunity Zone Equity	Peninsula parcels likely within Wilmington OZ boundary. OZ equity investors defer capital gains tax. Attracts private equity co-investors alongside grant stack.	\$1M–\$3M	OZ equity	Potential dilution — structure carefully
CDFI Equipment / Construction Loan	Reinvestment Fund, Artisans' Bank (DE-based), or NCALL Research. Manufacturing facility + equipment note at below-market rate. Ground lease as collateral support.	\$2M–\$4M	4–6%	15–20 yr term
Supplier Vendor Financing	CLT manufacturers (TerraLam, Structurlam, Mercer) sometimes carry equipment/inventory notes on anchor customer relationships. Part of the supply agreement negotiation.	\$500K–\$1.5M	Negotiated	Secured on equipment/inventory
M4D / Merge4Ward Equity	Principals' contribution. Minimized by grant stack. Ground lease eliminates land acquisition capital. Target:	\$250K–\$500K	0%	From operating cash flow if possible

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
	under \$500K personal exposure.			
M4CleanTech DE BOO (Klean/KBR)	Klean/KBR funds, owns, and operates the TyreNova facility on the southern tip parcel under BOO sublease from M4CleanTech. Zero capital requirement from Merge4Ward or M4D for the clean tech facility.	\$8M–\$15M (operator-funded)	BOO operator	No M4D capital required
TOTAL CAMPUS — All Sources	Manufacturing facility + clean tech BOO combined. M4CleanTech DE funded 100% by Klean/KBR.	\$13.75M–\$30.5M	Blended ~2–4%	M4D exposure: <\$500K

ROM ±30–40%. DEDO and EDA grant amounts depend on job creation commitments and competitive application quality. OZ equity requires fund formation or qualified investor identification — securities counsel required. CDFI loan availability depends on lender underwriting at time of application. Supplier vendor financing is subject to commercial negotiation.

SECTION 6 — CHESTER PA + WILMINGTON DE: THE TWO-NODE MID-ATLANTIC STRATEGY

Running Both in Parallel Amplifies Each Deal

Chester and Wilmington are not competing priorities — they are mutually reinforcing. Every step taken in Chester strengthens the Wilmington conversation, and vice versa. Trevor needs to present both to Mayor Roots and Sean Park as a coordinated regional strategy, not two separate asks.

	CHESTER PA (M4CLEANTECH + MAXWELLNESS)	WILMINGTON DE (MERGE4WARD + M4CLEANTECH)
Primary Use	Klean/KBR BOO tire pyrolysis (Phase 1) + Ecolomondo owned facility (Phase 2) + MaxWellness Community Health Campus	Merge4Ward CLT/SIP panel manufacturing + Klean/KBR BOO tire pyrolysis (Phase 1 DE node)
Ground Lease From	City of Chester / DCIDA	City of Wilmington — Sean Park / OED
Key Relationship	Mayor Stefan Roots + DCIDA	Sean Park (existing M4D relationship) + Mayor Purzycki
EJ / Grant Angle	Chester documented EJ community — EPA, EDA, DOE, PHFA all unlocked	Wilmington EJ overlay + Opportunity Zone + DEDO manufacturing incentives

M4D Revenue	M4CleanTech ground lease spread + developer fee + MaxWellness architecture fee	Merge4Ward manufacturing NOI + M4CleanTech DE ground lease spread + procurement savings
Chester Helps Wilmington	Chester BOO track record = Wilmington CDFI underwriting; Chester grant awards = Wilmington EDA competitiveness	—
Wilmington Helps Chester	Merge4Ward Wilmington node = Forward Building System supply for Chester MaxWellness build	—

THE COMPOUNDING ADVANTAGE TREVOR SHOULD ARTICULATE TO BOTH MAYORS

Chester and Wilmington are not two cities competing for M4D's attention — they are two anchor cities in a coordinated Mid-Atlantic clean industrial and affordable housing strategy. **Mayor Roots gets a clean tech anchor and a community health campus.** Mayor Purzycki gets a manufacturing node that supplies affordable housing panels to Delaware projects while co-locating a clean tech revenue generator. **Every building Merge4Ward delivers in Delaware uses panels manufactured on the Wilmington waterfront — from a City-owned ground-leased site — creating a closed loop of Delaware-sourced, Delaware-manufactured, Delaware-delivered community housing.**

SECTION 7 — IMMEDIATE ACTIONS FOR TREVOR

Three Authorizations. Two Cities. One Coordinated Ask.

NOW — Q3 2026 OED Parcel Meeting Sean Park — existing relationship Present Hay Road Peninsula aerial + campus concept to Sean Park at Wilmington OED. Request: (1) parcel ownership confirmation on 26-044/26-045/26-052 series, (2) City's interest in master development agreement, (3) intro to Mayor Purzycki for formal endorsement. Bring Chester brief as proof of parallel momentum.	Q3 2026 Supplier Pre-Qualification Ground lease as negotiating credential Initiate conversations with TerraLam, Structurlam, Acme Panel, and Aercon/Xella using the pending Wilmington ground lease as the anchor credential. Do not finalize terms before ground lease is signed — but establish relationships and get indicative pricing that reflects committed-site buyer status.	Q3–Q4 2026 Ground Lease + BOO Parallel Legal + Klean/KBR simultaneously Engage DE transactional counsel for ground lease negotiation. Simultaneously initiate Klean/KBR BOO site feasibility conversation for the 26-052 southern tip parcel. Right of First Refusal on adjacent parcels must be a ground lease term — negotiate this before signing.	Q4 2026 — 2027 Capital Stack Assembly Grant applications + CDFI File EPA Brownfields Assessment grant (Peninsula). File DEDO Strategic Fund pre-application. File EDA EAA pre-application citing Chester PA as parallel project. Engage Reinvestment Fund and Artisans' Bank for CDFI manufacturing loan pre-qualification. OZ investor identification begins.
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PREPARED BY & CONTACT

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All ROM investment, revenue, and cost-saving figures are preliminary estimates with ±30–40% confidence range. Not guaranteed, not funded commitments, and not GMP. Procurement savings estimates are based on published volume discount schedules and regional freight benchmarks — actual savings depend on negotiated supplier agreement terms and committed annual volumes. Ground lease terms, developer fee, BOO sublease economics, and supplier exclusivity arrangements all require negotiation and legal review before execution. CDFI and grant capital subject to competitive application and lender underwriting. Opportunity Zone investment requires qualified OZ fund and securities counsel review. DNREC environmental review and DEDO approvals are conditions precedent to any construction commitment.